

DESIGN AND CHARACTERIZATION OF VARIABLE ATTENUATORS AND PHASE SHIFTERS WITH LINEARIZED BIAS CONTROL VOLTAGES

M. Mtimat and A. B. Kouki

Department of Electrical Engineering,
École de technologie supérieure, 1100, Notre-Dame St. West,
Montréal (Québec), H3C 1K3, Canada
akouki@ele.etsmtl.ca

Abstract

In this paper, we first present the design, fabrication and test of a voltage controlled variable attenuator and a voltage controlled variable phase shifter. The designs use a reflection-type topology with PIN diodes for the attenuator and Varactor diodes for the phase shifter. Two measurement-based diode models are developed to optimize the design and insure first-pass success during fabrication. These models capture both DC and S-parameter behavior of the devices. Various impedance transformation topologies are investigated to maximize the dynamic ranges while minimizing phase-shift, for the attenuator, and insertion loss, for the phase shifter. The optimized designs are then fabricated and the linearization of the bias control voltage is investigated using both digital and analog approaches. The topologies of both configurations are presented and discussed and results of their implementations are presented.

Keywords: linear control, vector modulation, variable attenuator, variable phase shifter, linearization

1. INTRODUCTION

Microwave vector modulators using voltage-controlled variable attenuators [1-2] and variable phase shifters [3-4] are key components in many modern RF circuits and subsystems. Whether they are used for precise gain control, for modulation or as a part of complex signal cancellation circuits, such as feedforward amplifiers [5], it is desirable that these components exhibit linear behavior versus control voltage. This would facilitate their use as well as the implementation of precise and, easily programmable, control strategies in closed loop or open loop environments. With a linearized control voltage, direct linear vector modulation becomes

feasible at RF frequencies. This may then be used for such applications as MIMO (Multiple Input Multiple Output) channel emulation using time varying bias voltages [6].

The remainder of this paper is organized as follows: section 2 presents the reflective topology used for of the variable phase shifter and variable attenuator. Section 3 presents the design for the variable attenuator while section 4 presents the phase shifter design. In section 5 the bias linearization techniques are presented and discussed. Finally, concluding remarks are given.

2. GENERAL ATTENUATOR AND PHASE SHIFTER REFLECTIVE TOPOLOGY

The reflective type topology is used to design both the phase shifter and attenuator. This topology uses a 90° hybrid coupler, a four-port device, with identical variable loads, i.e., voltage controlled impedances, placed at its direct and coupled ports. The resulting circuit is a two-port device which can behave as a variable attenuator or a variable phase shifter depending on the nature of load impedance used. When the loads are identical and the hybrid coupler is well balanced, the input port is automatically matched and the output port, the coupler's isolated port, produces the output signal with the desired amplitude or phase modulation.

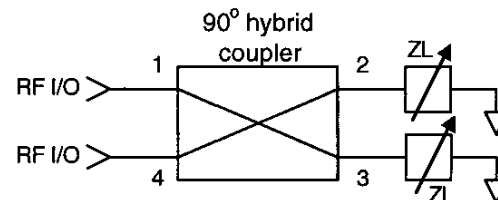


Fig. 1 Reflective topology for the attenuator and phase shifter.

The 90° hybrid coupler

At the heart of the proposed topology for realizing both the attenuator and phase shifter is the hybrid coupler. Since our interest is in frequencies around the PCS band, a commercial surface hybrid coupler for this band was chosen for this work.

3. ATTENUATOR DESIGN

To realize a variable attenuator using the topology shown in Fig. 1, the variable impedance must represent a variable resistance. This can be achieved using transistors or PIN diode elements. The HSMP3810 PIN diode, made by Agilent Technologies, was chosen for our application as it was readily available. Because the performance of the variable attenuator depends greatly on the diode, an initial step of the design was to construct a port circuit for modeling purposes. Fig. 2 shows the layout of one such circuit including the bias circuit.

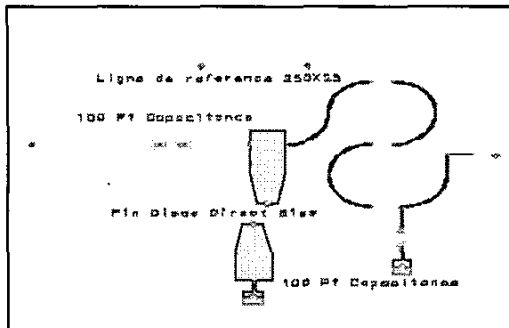


Fig. 2 PIN diode characterization circuit.

The results of initial measurements and modeling showed that the configuration shown in Fig. 2 would not provide sufficient attenuation range since it is not able to present a resistance close enough to 50 Ω . (see S22 of Fig. 3). To remedy this, a inductive transmission line impedance transformer was added rotating the impedance presented by the diode to the one by S11 in Fig. 3. Using this new configuration, the attenuator was fabricated and tested. The measured results for attenuation and phase-shift versus bias are shown in Fig. 4. The input and output matching was better than -15 dB for all bias conditions at the central frequency. Clearly, the attenuation versus bias voltage is highly nonlinear showing a very steep slope around a bias of 0.68V, which is the bias that yields a PIN resistance closest to 50 Ω .

4. PHASE SHIFTER DESIGN

To realize a variable phase shifter using the topology shown in Fig. 1, the variable load impedance must

represent a variable capacitor. This can be achieved by using varactor diodes. In this work, the VM34 diode, made by Metellics Inc., was chosen. A one-port circuit similar to that of Fig. 2 was used to test and model the diode. Because the original diode configuration as shown in Fig. 2 did not provide sufficient capacitance variation, hence limited phase shift, a new configuration that includes two varactors separated by a $\lambda/4$ line was chosen following [3] as shown in Fig. 5. Using this modified topology of the hybrid coupler originally chosen, the phase shifter was designed and fabricated. The measured response of the phase shifter as a function of the bias voltage is shown in Fig. 6 for different frequencies. A minimum phase shift range of 80° was obtained with less than 1 dB insertion loss and better than 26dB return loss at both ports. Again, the phase shift versus bias voltage exhibits a nonlinear behavior.

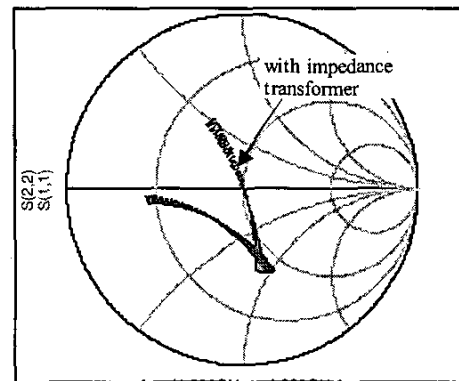


Fig. 3 PIN diode small signal response.

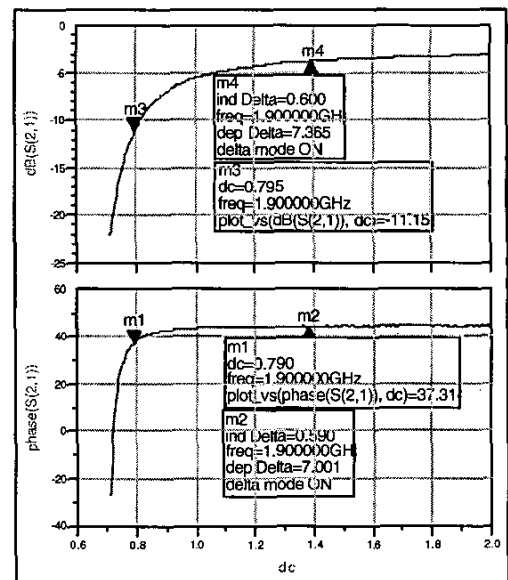


Fig. 4 Measured attenuator response versus bias

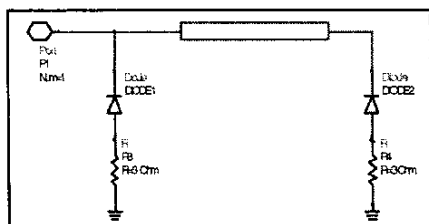


Fig. 5 Modified varactor diode configuration.

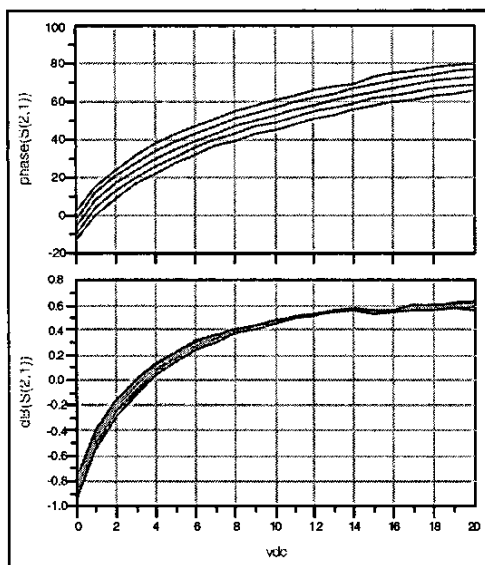


Fig. 6 Measured phase shifter response versus bias voltage.

5. LINEARIZATION OF THE RESPONSE VERSUS BIAS VOLTAGE

As shown in Figs. 4 and 6, the response of the variable attenuator and variable phase shifter versus bias voltage is nonlinear. In this manner, one does not have a direct and simple correspondence between the applied voltage and the resulting attenuation or phase shift. This in turn results in nonlinear behavior being introduced through the variable bias in applications that use these components. To remedy this problem, and to make the voltage-attenuation and voltage-phase shift responses linear, a proper linearization technique is required. To this end, a DC-bias transformation, or predistortion, circuit is required as shown in Fig. 7. The role of such a circuit is to produce an output bias voltage based on the input bias voltage and knowledge of the component's response. The transformed bias voltage would then yield a linear behavior of the component's response versus bias voltage. Such functionality may be implemented by either of two approaches: analog or digital.

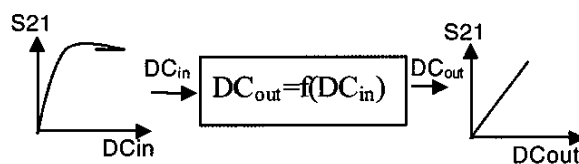


Fig. 7 Modified varactor diode configuration.

A digital implementation was performed using a standard microprocessor (ST500 from AMTEL). A lookup table was used to transform input bias voltage values to output values based on the data extracted from measurements. The lookup table was filled using values from a curve that was generated by a digital processor and can be implemented by a simple lookup table. The resulting measured attenuation versus predistorted bias voltage is shown in Fig. 8. A slope of 0.322 dB per 10 mV was obtained over a more than 10 dB attenuation range.

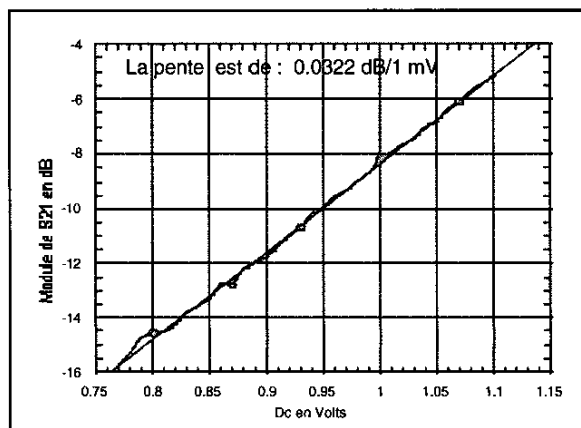


Fig. 8 Measured digitally linearized response of the attenuator.

A similarly linear response for the attenuator was obtained as shown in Fig. 9. The resulting slope is quasi-linear with a value of 4.49 degrees per V and overall linear phase shifting range of 60°. The ripples in the linear response for the phase shifter are due to the limited 8 bit resolution of processor used over the wide range of voltage needed to achieve the desired phase shift.

The digital linearization approach has been found to work reasonably well. However, it has significant limitations in that it requires a faster processor, modulates the bias voltages at sufficiently high speed, and higher bit resolution to achieve wider dynamic ranges. Furthermore, when the digital linearization circuit is operated near its maximum resolution it becomes quite sensitive to board level noise. An alternative to digital bias control linearization is to use an analog approach. In this approach, a curve fit of the desired DCout Vs. DCin response is plotted as shown in Fig. 10. Then, a nonlinear bias circuit is sought that would have a DC-DC transfer

characteristic as close to this response as possible. One possible such a circuit was consisted and was made of BJT transistor followed by an operational amplifier. Simulated results using this circuit and our variable attenuator are shown in Fig. 11.

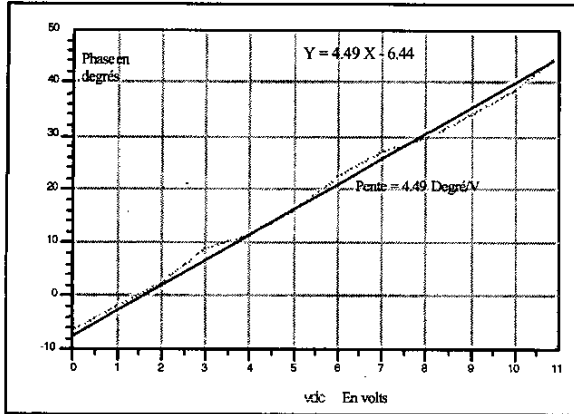


Fig. 9 Measured digitally linearized response of the attenuator.

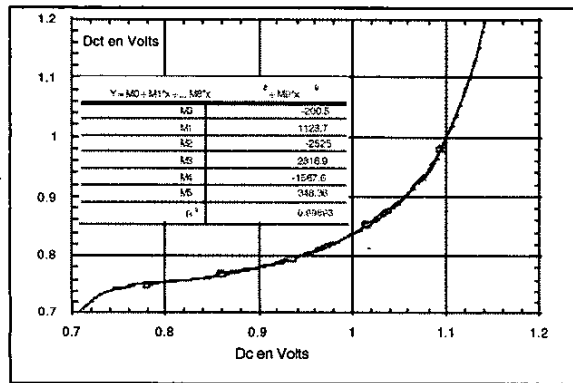


Fig. 10 Desired DCout Vs. DCin response.

5. CONCLUSION

In this paper we presented the design, characterization and bias linearization of a variable attenuator and a variable phase shifter. Attenuator and phase shifter circuits were fabricated and tested for the PCS frequency band. Two alternative bias linearization techniques were investigated. The first technique uses a digital approach and was implemented using a standard microprocessor. Good measured results were obtained showing a 10 dB linear attenuation range for the attenuator and a 60° linear phase shifting range for the phase shifter. The second technique, based on an analog approach, was shown to be feasible though the resulting linearity was not ideal. Further work is required on the analog approach to further improve it and validate it.

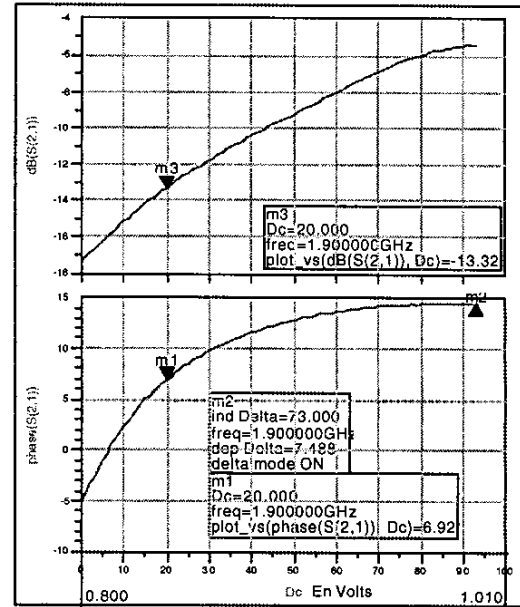


Fig. 11 Simulated bias control linearization using an analog approach.

Acknowledgements

This work is supported in part through NSERC funding.

References

- [1] C. W. Park, A. B. Kouki, F. M. Ghannouchi and E. A. Allamando, "A Variable Attenuator with Flat Group Delay Characteristics for PCS Applications", *28th European Microwave Conf.*, October 5-9, 1998, Amsterdam, Holland, Symp. Proceedings, pp. 684-688.
- [2] B. Maoz, "A novel, linear voltage variable MMIC attenuator," *Microwave Theory and Techniques, IEEE Transactions on*, vol. 38, pp. 1675-1683, 1990.
- [3] C. W. Park, A. B. Kouki, F. M. Ghannouchi and E. A. Allamando, "Analog 360 Degrees Phase Shifters for PCS Applications", *1998 IEEE MTT-S European Wireless 98*, October 8-9, 1998, Amsterdam, Holland, Symp. Proceedings, pp. 319-324.
- [4] H. N. Hayashi, T.; Araki, K., "A miniaturized MMIC analog phase shifter using two quarter-wave-length transmission lines," *Microwave Theory and Techniques, IEEE Transactions on*, vol. 50, pp. 150-154, 2002.
- [5] G. Zhao, F. M. Ghannouchi, F. Beaugard and A. B. Kouki, "Digital implementations of adaptive feedforward amplifier linearization techniques," *IEEE MTT-S International Microwave Symposium Digest*, Vol. 2, pp. 543-546, June 1996.
- [6] P. P. Carpentier, A. B. Kouki and C. Thibault, "A flexible digital platform for real-time control of RF components with application to MIMO channel emulation," *IEEE Canadian Conference on Electrical and Computer Engineering*, Niagara-Falls, May 2002.